

RESEARCH INTERESTS

Mathematical Physics, Quantum Field Theories, Quantum Foundations

I am keenly interested in using mathematical structures to glean insights into physical theories and am particularly interested in working in mathematics that may lead to insights into quantum foundations, quantum gravity, or string theory. While I don't have any particular question in mind, I would like to work on a project that would let me experience mathematical physics research. Currently, I am reading up on Yang-Mills theory, and learning Conformal Field Theory.

EXPERIENCE

- Attended RADCOR 2025 - 17th International Symposium on Radiative Corrections: Applications of Quantum Field Theory to Phenomenology
<https://indico.niser.ac.in/event/2/> Oct 2025
- Mathematical Foundations of Quantum Mechanics
Guide: Anupam Pal Choudhury, School of Mathematical Sciences, NISER Jan 2026 - Present
Operator Theory, results for unbounded operators and infinite dimensional Hilbert spaces, spectral decomposition.
- Quantum Field Theory
Guide: Shamik Banerjee, Schools of Physical Sciences, NISER Feb 2025 - Present
Scalar ϕ^4 theory, QED, renormalisation, RG flow, Yang-Mills Theory, CFT
- Formalisms in Classical and Quantum Mechanics
Guide: Sujay K. Ashok, Dept. of Theoretical Physics, IMSc May 2024 - Jan 2025
Canonical transformations in Hamiltonian mechanics; symplectic transformations; Poisson bracket formulation; path integral formalism in QM; resolvent, green's function and quantum scattering; homology groups, Lefschetz thimbles and Borel resummation.
- Non-Linear Dynamics and Chaos
Guide: Sayantani Bhattacharya, School of Physical Sciences, NISER Summer 2023
Bifurcations, hysteresis and other phenomena in 2-D systems, rudimentary analysis of higher dimensional non-linear systems. Numerical analysis of the parameter space of the Lorenz Equations.
- Linear Dynamical Systems
Guide: Anupam Pal Choudhury, School of Mathematical Sciences, NISER Summer 2023
An initial but rigorous foray into linear dynamical systems and bifurcations.

PROJECTS

- Defining a Metric on the Set of Images
- Percolation; Random Walks and Diffusion
- Exercises from Peskin and Schroeder QFT book
- Spherical model of a ferromagnet
- Magnon Bound States in 1d Heisenberg Chain
- Bead on a Inclined Rotating Hoop, Chaos and Resonances

ACHIEVEMENTS

- JEST-2026 exam **All India Rank 9**
- GATE-2025 exam **All India Rank 135**
- Batch-22 Topper in the first semester, NISER
- Innovation in Science Pursuit for Inspired Research (INSPIRE), Scholarship issued by Department of Science and Technology

SKILLS

- **Relevant Topics that I have studied:**
 - Functional Analysis
 - Manifolds Theory and Differential Geometry
 - Measure Theory and Integration
 - Phase Transitions and Critical Phenomena
 - Quantum Field Theory and RG flows
 - General Theory of Relativity
 - Quantum Information and Computing
 - Quantum Optics
 - Programming (Embedded, and Numerical)
- **Currently learning:**
 - Non-Abelian Gauge Theory
 - Conformal Field Theory

EDUCATION

National Institute of Science Education and Research, Integrated BSc-MSc (A five year undergraduate cum master's course) – Currently finished eighth semester – CGPA: 9.23	Bhubaneswar, Odisha, India 2022–Current
Chavara Public School CBSE Class XII – 96.4% with Physics, Mathematics, Computer Science, Chemistry	Kottayam, Kerala, India 2022
St. Jude's Global School CBSE Class X – 96.6%	Kottayam, Kerala, India 2020

TALKS AND SESSIONS

- “*A Physicist's Introduction to Lie Groups and Algebras*” - NISER Physics Club, Nov 2025
 - “*The N-Queens Problem*” - NISER Maths Club, 2022
 - Informal tutoring for juniors in electromagnetism, nuclear physics, and particle physics.
- ## EXTRACURRICULAR ACTIVITIES
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- Former Head of Coding Club, NISER (2024-26)
 - Former Core Comittee Member of Physics Club, NISER (2023-24)
 - Former Head of Chess Club, NISER (2024-25)
FIDE rating of 1470.
 - Piano
Merit in Grade 5 Exam, Trinity School of Music, London.
 - GM of a Dungeons & Dragons Campaign.